Fiscal Year:	FY 2024	**Task Last Updated:**	FY 01/10/2023
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Round Robin - Thermophysical Property Measurement		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	04/01/2017	**End Date:**	02/28/2023
No. of Post Docs:	3	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	9	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 02/28/2023 per PI (Ed., 1/10/23). NOTE: End date changed to 02/29/2024 per PI (Ed., 4/4/22).		
Key Personnel Changes/Previous PI:	Co-Investigator Mikhail Krivilev from Udmurt State University has been dropped from the program due to inactivity on the grant. He was unpaid by NASA since he is supported by ROSCOSMOS, so no fiscal impact.		
COI Name (Institution):			
Grant/Contract No.:	NNX17AH41G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Researchers go to space to conduct material property measurements that are required to model terrestrial manufacturing processes to provide higher performance and discover new materials to make people's lives better around the world. We justify the expense of doing tests in microgravity by concentrating on those materials which are impossible to evaluate on Earth or have properties that can only be measured with higher precision when gravitational accelerations are minimized. We know we get better results but we do not yet know how much better. The JAXA (Japanese Aerospace Exploration Agency) Electrostatic Levitation Furnace (ELF) was designed to process high temperature oxides but it can be readily adapted to allow testing of industrially important metals and alloys as well. By selecting materials that are already being processed using another technique, the European Space Agency (ESA) Electromagnetic Levitation facility on the International Space Station (ISS), these two facilities can be compared so that researchers know their relative strengths such that they may complement each other to maximize science output from space tests. Our objectives are to conduct space experiments to compare to ground results on a baseline material set and to model the influence of gravity on measurement techniques by various research groups across the world. Our approach is to employ Round Robin testing by multiple researchers on the same material using different techniques, on different facilities, with different analysis protocols to compare results. The significance of this work is that we will then know the relative increase in measurement fidelity and which material works best in which facility to efficiently use the limited available space sciences resources such as launch cost, crew time, and telemetry bandwidth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This program leverages the existing ESA International Topical Team organization to develop a strategy to compare the performance of current flight and ground facilities investigating thermophysical property measurement using levitation techniques. New experiments are proposed for the JAXA-ELF facility on the ISS to complement other ongoing investigations that use both the ISS-based Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) and ground-based facilities. We will efficiently use available resources, both fiscal and crew time, by selecting materials compatible with those previously prioritized by other existing research groups. In doing so, we will expand the capabilities of space electrostatic levitation processing to allow for investigation of additional thermophysical material properties and manifesting new material classes across microgravity platforms. The key scientific questions we address are: (1) How do we control and minimize gravity-induced systematic error in thermophysical property measurement and does theory explain the observed deviations in results? (2) For which material classes are space-based platforms delivering thermophysical property measurement results with higher accuracy and/or precision than ground-based facilities? Thermophysical property testing in microgravity provides enhanced fidelity of results purportedly due to three main effects. First, without strong gravitational accelerations and with reduced levitation forces a more spherical sample allows for better analysis of experimental behavior. Second, better control of convection in space results in higher measurement precision. Third, sedimentation and buoyancy induced segregation are eliminated in reduced gravity. But how much better do we do in space as compared to ground? The obvious solution is to quantify variability across platforms – both in space and on ground.
Task Progress:	This report serves as a final project summary report, and details the thermophyical property testing on metallic melts using the Japanese Space Agency (JAXA) electrostatic levitation furnace (ELF) and NASA Marshall Space Flight Center (NASA-MSFC) electrostatic levitation facilities. It leveraged the existing European Space Agency (ESA) International Topical Team organization and an international consortium of researchers across the globe to develop a strategical performance comparison of current flight and ground facilities investigating thermophysical property measurement using levitation techniques. This report includes results from the study of four key thermophysical properties on three pure metals Au, Zr and, Pt. Density and thermal expansion coefficient have been measured as paired properties for all three metals. The novel "Frequency crossover" natural frequency identification technique was developed during this program and applied during surface tension and viscosity measurement to complement the traditional "Maximum Amplitude" method previously developed to support analysis of Faraday-forced excitation of sample deformation. Working in tandem, this provides post-test validation of the quality of a particular experiment. Surface tension and viscosity have been measured as paired properties from the confirmed natural frequency oscillations. The ratio of specific heat capacity and emissivity for all three metals are also being reported as a part of this study. An attempt has been made to quantify the possible sources of uncertainties associated with measured properties during this project using the GUM (Guide to the Expression of Uncertainty in Measurement) principles. These measured uncertainties have been used to develop a performance quantification method, called a Measurement of Accuracy and Precision (MAP) diagram,which allows direct comparison of facility accuracy and precision. This enables future researchers to assess the reliability of the reported values and better understand the advantages and challenges associated with each facility.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Azersky C, Jeon S, Settens C, Fakhrul M, Matson D. "Quantifying X-ray diffractometer precision for measurement of dislocation density and the evolution of defect structures during solidification of additive manufacturing powders." IOP Conf. Ser.: Mater. Sci. Eng. 2023;1281:012033. http://dx.doi.org/10.1088/1757-899X/1281/1/012033 , Jun-2023
Articles in Peer-reviewed Journals	Nawer J, Stanford B, Kolbe M, Schneider S, Gossé S, Wunderlich RK, Mohr M, Borzì A, Neels A, Matson DM. "Thermodynamic assessment of evaporation during molten steel testing onboard the International Space Station." npj Microgravity. 2024 Jul 19;10(1):77. https://doi.org/10.1038/s41526-024-00416-1 , Jul-2024
Articles in Peer-reviewed Journals	Nawer J, Ishikawa T, Oda H, Koyama C, Matson DM. "Uncertainty quantification of thermophysical property measurement in space and on earth: A study of liquid Platinum using electrostatic levitation." J Astron Space Sci. 2023 Sep 15;40(3):93-100. https://doi.org/10.5140/JASS.2023.40.3.93 , Sep-2023
Articles in Peer-reviewed Journals	Muusha P, Stanford B, Kolbe M, Matson DM. "Relating cooling rates in superheated liquid during solidification for powder characterization." Metall Mater Trans A. 2023 Nov;54(11):4169-78. https://doi.org/10.1007/s11661-023-07135-4 , Nov-2024

Articles in Peer-reviewed Journals	Nawer J, Ishikawa T, Oda H, Koyama C, Saruwatari H, Sansoucie MP, Philips B, Gossé S, Mohr M, Kolbe M, Matson DM. "A quantitative comparison of thermophysical property measurement of CMSX-4® Plus (SLS) in microgravity and terrestrial environments." High Temperatures-High Pressures. 2023 May 1;52:323-39. http://dx.doi.org/10.32908/hthp.v52.1407 , May-2023